BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 12 READING MATERIAL

Interpretation & Writing I

From Plots to Text

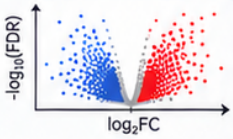
Reading papers, evaluating evidence, and writing scientific results

Multi-Omics Evidence

Genomics / Epigenomics



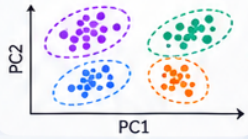
Transcriptomics



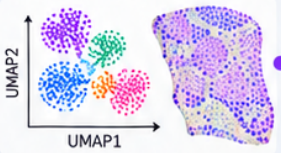
Proteomics / Metabolomics



Microbiome



Single-Cell / Spatial



From Evidence to Writing

1. Scientific Question



Identify the gap and hypothesis.

2. Study Design



Check samples, controls, and comparisons.

3. Figure Interpretation



Read patterns, effects, and uncertainty.

4. Evidence Assessment



Examine alternatives and experimental support.

5. Scientific Argument



Connect findings to supported claims.

6. Results & Legends



Write clear, traceable conclusions.

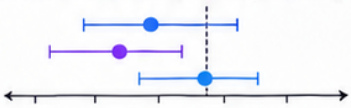
Claim Strength

Observations • associations • causal claims



Statistical Evidence

Effect size • uncertainty • multiple testing



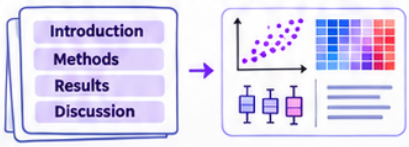
Experimental Support

Independent samples • perturbation • rescue



Manuscript Structure

Introduction • Methods • Results • Discussion





CRITICAL READING

Identify the question and evaluate the design.

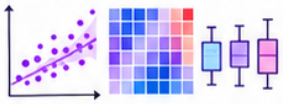
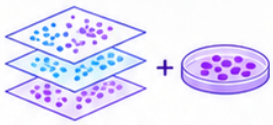


FIGURE INTERPRETATION

Explain the measurements and their limits.



EVIDENCE INTEGRATION

Connect omics findings with experiments.



SCIENTIFIC WRITING

Turn supported findings into results and legends.

Fall 2026 | Two 40-Minute Undergraduate Sessions | Dr. Liwei Xie

How to use this chapter

A bioinformatics paper connects a biological question to measurements, analyses, and conclusions. As a reader, you evaluate whether those conclusions follow from the evidence. As a writer, you make that connection clear enough for another person to inspect, reproduce, and challenge.

This chapter brings together the analyses studied earlier in the course: genomics, bulk RNA-seq, microbiome analysis, proteomics, metabolomics, single-cell and single-nucleus transcriptomics, spatial transcriptomics, enrichment analysis, multi-omics integration, and data visualization. Epigenomic analysis is included to connect the regulatory examples discussed in the course. Computational analysis and laboratory experiments are considered together throughout.

The reading strategies apply broadly to biological discovery papers using bioinformatics. For computational methods papers, the same questions about evidence apply, but the central evidence may be benchmarking, ablation studies, robustness, and independent evaluation. A laboratory experiment is necessary when the claim requires it; it is not a universal requirement for every computational contribution.

Week 12 develops scientific interpretation and the first manuscript draft, with particular attention to Results paragraphs and figure legends. Week 13 develops AI-assisted polishing and critique. This progression follows the supplied course syllabus, which also connects multi-omics analysis to a final mini-paper containing Abstract, Methods, Results, and Discussion sections. [S1]

Learning outcomes

By the end of this chapter, you should be able to:

1. Reconstruct a paper's biological question, knowledge gap, study design, and central claim.
2. Interpret familiar bioinformatics figures while separating measurement, statistical evidence, and biological explanation.
3. Identify the independent biological unit, the relevant comparison, and important sources of uncertainty.
4. Explain how computational findings, independent replication, and laboratory experiments contribute different kinds of evidence.
5. Write connected Results paragraphs and a standalone figure legend without overstating the findings.
6. Assemble a manuscript outline in which each major claim can be traced to a figure, analysis, or experiment.

Reading map

The allocation below is a proposed structure for expanding this framework into an approximately 32-page handout, with references as needed. It is not the pagination of this framework file. Read Sections 1–3 and 5–8 as the core chapter. Use Section 4 as a reference atlas, then complete the case reading and writing tasks in Sections 9–10.

Section	Main question	Suggested space
Orientation	What will you learn and produce?	1 page
1. Read a paper as a scientific argument	What question does the paper answer, and how?	3 pages
2. Understand the design and statistical evidence	What was compared, and what makes the comparison credible?	3 pages
3. Read any figure systematically	What is measured, displayed, and supported?	2 pages
4. An interpretation atlas across omics	How do the same questions apply to each analysis family?	8 pages
5. Connect computational and experimental evidence	What does each additional experiment establish?	3 pages
6. Turn figures into Results paragraphs	How do you write a precise, connected account of the findings?	3 pages
7. Write legends and reproducible Methods	What must another reader know to interpret and reproduce the result?	2 pages
8. Assemble the manuscript	How do the sections work together?	2 pages
9. Guided reading of integrated studies	How do published papers build and qualify their arguments?	2 pages
10. Practice and your writing portfolio	Can you apply the framework to your final project?	3 pages

1. Read a paper as a scientific argument

1.1 Separate the topic, question, hypothesis, and conclusion

The topic identifies the broad area. The question specifies what remains unknown. A hypothesis offers a testable explanation. A conclusion states what the study's evidence supports.

For example, in an illustrative study of muscle regeneration:

- **Topic:** Age-related changes in muscle regeneration.
- **Question:** Does altered chromatin regulation contribute to impaired stem-cell differentiation?
- **Hypothesis:** A regulatory change modifies a transcriptional program needed for differentiation.
- **Alternative explanation:** Differences in the measured population reflect cell composition or survival rather than a change within otherwise comparable cells.
- **Conclusion:** Its strength depends on the actual design and results. An expression difference, a regulatory association, and a demonstrated functional contribution require different evidence.

When reading the Introduction, write down what was known, what remained uncertain, why that uncertainty matters, and which experiment could distinguish the competing explanations.

1.2 Read with a purpose

Use three adaptable passes. Active reading should be guided by what you need to understand, rather than by a fixed rule about which section must always come first. [1]

Pass 1 — Establish the question. Read the title, abstract, relevant background, and figure overview. Write a provisional statement of the authors' question and answer.

Pass 2 — Reconstruct the evidence. Read each important figure together with its legend, the relevant Results paragraph, and the Methods needed to understand the comparison. Record which samples generated each result.

Pass 3 — Test the argument. Examine controls, supplementary results, alternative explanations, exclusions, sensitivity analyses, and data availability. Revisit the abstract and ask whether its strongest statement is justified.

Do not postpone a necessary Methods check. You cannot interpret a group comparison correctly if you do not know what constitutes a sample, whether samples were paired, or how the reported quantity was calculated.

1.3 Build an evidence map

For each major figure, complete one row in a table with the following fields:

Field	What you record
Biological question	The specific uncertainty addressed by this figure
Panel and source	Figure number, panel label, and data source
Design and measurement	Biological system, independent sample unit, groups, and measured quantity

Field	What you record
Main observation	Direction and magnitude of the finding, with uncertainty where available
Statistical evidence	Model or test, adjustment, and relevant assumptions
Supported claim	The strongest statement justified by this evidence
Remaining explanation	A plausible alternative not resolved by the panel
Next test	An analysis or experiment that would distinguish the explanations

Keep the authors' claim separate from your assessment of its support. If a required detail is unavailable, record “not reported” rather than guessing.

Practice: Summarize a paper in five sentences: background, gap, approach, principal finding, and remaining uncertainty.

2. Understand the design and statistical evidence

2.1 Identify the independent biological unit

Before interpreting any significance label, identify what was independently sampled or assigned to a condition. Depending on the design, this might be a participant, animal, independently established culture, or another experimental unit.

Distinguish biological replicates from repeated observations and technical measurements. Thousands of cells from one donor provide information about that donor; they do not create thousands of independent donors. Single-cell differential expression that ignores variation between biological replicates can generate misleading discoveries. [3]

For each paper, check:

- How many independent units were included in each group?
- Which observations were paired, repeated, nested, or pooled?
- Were RNA, protein, metabolites, and other measurements obtained from the same biological samples?
- Were discovery and validation samples independent?
- Were treatment, sequencing batch, collection site, sex, age, or other variables confounded?
- Were exclusion criteria, randomization, and blinding appropriate and reported?

A second omics measurement from the same samples provides another molecular view. It is not automatically an independent biological replication. Likewise, a new analysis of the same public dataset does not become an external validation simply because it uses different software.

2.2 Read the effect before the significance label

Ask how large the difference is, in which direction it occurs, and how precisely it was estimated. Then evaluate the statistical evidence under the stated model.

A P value does not give the probability that the hypothesis is true, and statistical significance does not establish biological importance. An estimate with a wide confidence interval may remain compatible with several biologically different interpretations. [4]

For an omics result, record the comparison, effect-size scale, uncertainty, and multiplicity treatment. Distinguish raw P values, adjusted P values, and the stated false-discovery-rate procedure. FDR concerns the expected proportion of false discoveries among discoveries under the method's assumptions; it is not a per-feature probability that an individual biological claim is false.

Do not equate a nonsignificant result with proof of no effect. Ask whether the study had enough precision to exclude effects that would matter. Do not claim that two groups respond differently merely because one within-group test is significant and the other is not; the difference in responses needs its own comparison.

2.3 Check whether processing choices affect the conclusion

Inspect the rationale for filtering, normalization, missing-value handling, batch adjustment, covariate inclusion, and feature selection. Ask whether the conclusion survives reasonable alternatives. If every case was measured in one batch and every control in another, software cannot fully separate the biological effect from the batch effect using those data alone.

Practice: Draw a design map for a study with paired RNA-seq and proteomics measurements. Mark independent samples, repeated measurements, processing batches, and the contrast used in the statistical model.

3. Read any figure systematically

3.1 Use eight questions for every panel

1. **Question:** What uncertainty does the panel address?
2. **Measurement:** What was directly measured, and what was inferred?
3. **Comparison:** Which conditions, cells, features, or regions are compared?
4. **Display:** What do the axes, scales, points, colors, shapes, and error bars mean?
5. **Evidence:** What are the effect size, sample structure, uncertainty, and statistical support?
6. **Interpretation:** What biological statement follows from the result?
7. **Boundary:** What stronger statement remains unsupported?
8. **Next test:** What would distinguish the remaining explanations?

Write an observation sentence before writing an interpretation sentence. For example, “The gene had higher estimated transcript abundance in the treated samples” describes a measurement. “The gene mediates treatment-induced regeneration” requires additional evidence.

3.2 Connect visual reading to the ggplot2 concepts you already know

Identify the data represented by each graphical mark and the variables mapped to position, color, fill, size, and shape. Determine whether the plot displays raw observations, a transformed quantity, a model estimate, or a summary calculated across observations.

Check the scale before comparing magnitudes. A row-scaled heatmap shows a gene's pattern relative to its own values; its colors do not directly compare absolute abundance between different genes. A

violin shows a distribution estimate, while a dot, box, or error bar may summarize a different aspect of those data. A displayed significance annotation must be traced to its underlying test.

For every figure, distinguish three types of work: the biological measurement, the statistical analysis, and the graphical display. A polished graphic does not resolve a weak design or an inappropriate model.

Practice: Annotate one volcano plot, one heatmap, and one PCA plot. For each, label the unit represented by a point or cell, the transformation used, and one conclusion the plot cannot establish by itself.

4. An interpretation atlas across omics

Use the same entry structure for each analysis family: **biological question; common outputs; essential checks; defensible wording; unsupported extension; useful validation.** The examples below are writing models, not reported findings from a particular study.

4.1 Genomics and functional annotation

Question: Which sequence differences or genomic features are associated with a phenotype, or could plausibly contribute to a function?

Read coverage and quality summaries, variant tables, association plots, genomic tracks, and gene or gene-cluster annotations. Distinguish a variant call from an association and an association from functional causality. Check the reference genome, filtering, population structure where relevant, and the method used to assign a locus to a gene.

Defensible wording: “The locus was associated with the phenotype, and gene X was prioritized for functional evaluation.”

For microbial genomes, annotation of a putative enzyme or pathway establishes a candidate function. It does not demonstrate substrate conversion, expression under the experimental conditions, or a strain-level phenotype. Targeted genetic and biochemical tests address those questions.

4.2 Bulk RNA-seq and differential expression

Question: Which transcripts differ between specified conditions, after accounting for the study design?

Read sample quality summaries, PCA, sample correlations, MA plots, volcano plots, heatmaps, and gene-level estimates. Check the reference condition, the sign of the contrast, biological replication, covariates, count filtering, and multiple-testing procedure. DESeq2 models count data; values prepared for visualization and values supplied to differential-expression modeling have different roles. [5]

Trace the result from aligned or quantified reads to the gene-level count matrix and then to the modeled comparison. Alignment alone does not identify differential expression.

Defensible wording: “Gene X showed higher estimated transcript abundance in treatment than in control after adjustment for the specified covariates.”

A volcano plot prioritizes statistical and quantitative changes, not necessarily causal regulators. A change in bulk expression may reflect altered cell composition, regulation within a cell type, or both.

Follow-up could involve cell-resolved analysis, independent samples, protein measurements, and a functional perturbation selected for the claim.

4.3 Epigenomics and regulatory interpretation

Question: Which regulatory properties of chromatin differ, and how are they related to transcription?

Distinguish what the assays measure: ATAC-seq profiles accessibility; transcription-factor ChIP-seq or CUT&RUN profiles enrichment associated with a selected protein; histone-mark assays profile the selected chromatin modification. H3K4me3, H3K9me3, H3K27me3, and H3K27ac should not be treated as interchangeable readouts or as deterministic labels for gene expression.

Read differential-peak results, genome-browser tracks, motif enrichments, peak annotations, and aggregate profiles around transcription start sites. ChIPseeker provides genomic annotation of peaks; deepTools supports signal summaries and visualization. Neither annotation nor an aggregate heatmap itself tests whether a particular regulatory element controls a gene. [13,14]

Defensible wording: “A region linked to gene X showed increased accessibility alongside higher transcript abundance, supporting a regulatory hypothesis.”

Check peak-to-gene assignment, signal normalization, controls, replicate consistency, and whether the compared cell states match. A motif is a sequence pattern; occupancy evidence, functional regulation, and direct sequence-specific action require additional, distinct support. Candidate tests include protein-occupancy assays, regulatory-element perturbation, and appropriately controlled reporter experiments.

Distinguish peak calling from differential accessibility or binding analysis. A peak detected in one condition but not another is not, by itself, a statistical test of their difference. Examine quantified signal across defined regions and biological replicates, and inspect the background used for motif enrichment, including analyses performed with HOMER.

4.4 Proteomics and protein function

Question: Which proteins or protein modifications differ, and what functional interpretation is justified?

Read identification and quantification quality, missingness, volcano plots, heatmaps, protein-level estimates, and modification-site results where relevant. Distinguish peptide evidence from protein inference and identification error control from differential-abundance testing. Check whether missing-value treatment could influence the apparent difference.

Defensible wording: “Protein X abundance increased, consistent with the RNA-level change.”

Concordant RNA and protein measurements strengthen an abundance-based interpretation. Protein abundance alone does not measure enzyme activity, localization, or physical interaction. Co-enrichment in an affinity-purification experiment does not by itself establish direct binding.

Practice: Explain why RNA and protein measurements might disagree. Consider timing, synthesis, degradation, isoforms, measurement precision, and cell composition before selecting a biological explanation.

4.5 Metabolomics and metabolic interpretation

Question: Which metabolite features differ, and what do those measurements reveal about metabolism?

Read quality-control plots, batch structure, PCA or PLS-DA, feature intensities, identified-metabolite tables, and pathway summaries. Distinguish an untargeted feature from a confidently identified compound, and relative signal from absolute concentration.

Defensible wording: "The measured abundance of metabolite M was lower in the treated samples."

A metabolite pool size depends on production, consumption, transport, and compartmentation. A concentration difference alone cannot establish the direction of metabolic flux or the organism that produced the metabolite. Select targeted measurements, isotope tracing, enzyme assays, or source-tracing experiments according to the proposed explanation. Kang et al. provide an optional case connecting metabolite measurements with chromatin and functional experiments. [15]

4.6 Microbiome composition and function

Question: How do community composition, diversity, or functional potential relate to the host or environment?

Read alpha-diversity measures, beta-diversity ordinations, taxonomic profiles, differential-abundance results, and metagenomic functional annotations. Identify whether the data came from 16S, 18S, ITS, or shotgun metagenomics and whether the taxonomic resolution supports the named organism.

Relative-abundance measurements are constrained by composition; a larger fraction does not automatically mean more organisms in absolute terms. [8] Check confounding by diet, medication, collection conditions, and host characteristics. For PERMANOVA, inspect the distance metric, design, permutation scheme, and group dispersion before interpreting separation.

Defensible wording: "Community composition differed between the groups under the specified distance-based analysis."

Higher diversity is not universally synonymous with better health. Predicted function from amplicon data differs from measured metagenomic potential, and neither alone demonstrates metabolic activity. Metabolite measurements, culture experiments, microbial genetics, and host interventions test different links in the proposed mechanism.

4.7 Single-cell and single-nucleus transcriptomics

Question: Which cell populations or states differ, and where do molecular changes occur?

Read quality filtering, doublet assessment, UMAP or t-SNE, marker plots, cell proportions, within-cell-type differential expression, and trajectory results. Distinguish cell identity from state and a change in cell abundance from a change in expression within a cell type. Single-nucleus and single-cell protocols can sample different RNA compartments and cell populations.

Defensible wording: "The analysis identified a treatment-associated expression difference within the annotated cell population."

Inspect donor-level evidence and an appropriate model for the sampling structure. Pseudobulk approaches aggregate within cell type and biological replicate; other models can also account for biological replication. [3] UMAP separation is not a significance test. Pseudotime describes an inferred ordering and does not, by itself, demonstrate chronological progression or lineage ancestry.

4.8 Spatial transcriptomics and inferred communication

Question: Where are cell states or molecular programs located, and which interactions are plausible?

Read tissue images, spot or cell maps, spatial expression, deconvolution results, neighborhood analyses, and ligand–receptor predictions. Check spatial resolution, the number of biological specimens, mixed-cell spots, tissue alignment, and uncertainty in annotation or deconvolution.

Defensible wording: "The two populations were spatially adjacent, and expression-based analysis nominated a candidate signaling interaction."

Co-location and ligand–receptor expression do not establish that signaling occurred or caused the phenotype. A stronger claim may require localization of relevant molecules, receptor or ligand perturbation, suitable coculture experiments, and a defined downstream response. Multiple tissue sections from one animal do not automatically provide independent animal replication.

4.9 Enrichment, networks, and multi-omics integration

Question: Which biological programs organize the molecular results, and what information is added by combining measurements?

For **over-representation analysis (ORA)** using GO or KEGG, identify the selected list, eligible background, annotation database, and multiple-testing correction. Distinguish GO biological process, molecular function, and cellular component categories. Read gene counts, gene ratios, the stated enrichment measure, and adjusted P values according to the plot's definitions; similar terms may share many of the same genes.

For **GSEA**, identify the ranking statistic, its direction, gene-set definition, normalized enrichment score, statistical procedure, and leading-edge features. Enrichment concerns a pattern in the measured or selected features; it is not a direct assay of pathway activity. [6]

Distinguish the purposes of common integration methods. PCA summarizes variation without using outcome labels. PLS-DA and DIABLO use group labels for discrimination. MOFA estimates latent sources of variation across modalities. Correlation networks represent statistical relationships whose biological meaning depends on design and assumptions. [9,10]

Defensible wording: "The omics layers shared a pattern associated with the phenotype, which prioritized a candidate process for validation."

Ask whether samples are matched, feature identifiers are compatible, shared batch effects remain, and the conclusion is stable. A Venn-diagram overlap is descriptive unless tested against an appropriate background. Agreement across layers does not establish causal order; disagreement may reflect timing, regulation, measurement coverage, or bias.

4.10 Predictive models and AI-assisted analysis

Question: Does a model predict an outcome in data that were not used to construct or select it?

Read ROC and precision–recall curves, calibration plots, confusion matrices, uncertainty estimates, and performance in independent datasets. Distinguish discrimination from calibration, and predictive performance from biological mechanism.

Check whether related samples cross training and test boundaries. Preprocessing learned from data, feature selection, and tuning must respect the evaluation split. Using outcome information from the test set during development can produce optimistic estimates. [11]

Defensible wording: “The model showed the reported predictive performance in the specified independent evaluation set.”

Feature importance does not establish causality. For AI-assisted interpretation, require traceable outputs and sources: an LLM-generated explanation is a draft to verify, not an additional biological measurement. Record the dataset, comparison, analysis version, result table, and figure supporting each manuscript statement.

5. Connect computational and experimental evidence

5.1 Ask what the additional experiment contributes

Avoid using “validation” as a single undifferentiated label. Different experiments answer different questions.

Evidence type	Question it addresses	Boundary to retain
Orthogonal measurement	Does another assay support the measured change?	Same-sample agreement is not independent biological replication.
Independent biological replication	Does the finding reproduce in new biological material?	Replication of an association does not establish its mechanism.
Loss of function	Is the candidate required for the response in this setting?	Consider specificity, viability, compensation, and off-target effects.
Gain of function	Can increasing the candidate produce the response?	Dose, localization, and physiological relevance constrain interpretation.
Rescue	Does restoring a component recover the response?	Recovery supports a particular relationship; it does not establish every proposed molecular step.
Mechanism-specific test	Does the proposed interaction, regulatory element, biochemical reaction, or causal order operate?	The assay and controls must address the precise mechanistic claim.
External predictive evaluation	Does a model generalize to the intended population or setting?	Predictive success does not itself establish a molecular mechanism.

5.2 Follow two recurring routes through a paper

Route A — Transcriptome, proteome, and function. Start with the RNA-level change. Determine whether protein measurements support the relevant molecular interpretation. Then ask whether an assay measures the proposed biological consequence and whether perturbing the candidate changes that consequence. Preserve discordant results and distinguish whole-pathway behavior from an individual molecular measurement.

Route B — Epigenome, transcriptome, and regulation. Start with the changed genomic region. Examine the evidence linking that region to a gene, the gene's expression response, and the proposed regulator. Ask which experiment tests regulation at the endogenous locus. Agreement between a

peak and an RNA change can prioritize a hypothesis without establishing the direction or directness of regulation.

The same reasoning transfers to microbiome–metabolome–host studies and single-cell–spatial studies. Every arrow in a mechanistic model should have an identifiable evidence source; unresolved arrows remain hypotheses.

5.3 Design a discriminating experiment

For one unresolved claim, specify the biological system, independent units, intervention, negative and positive controls where appropriate, primary readout, and alternative outcomes. Explain which outcome would favor each hypothesis and what result would remain ambiguous.

Practice: A candidate gene correlates with a phenotype. Propose one experiment that confirms the measurement and a different experiment that tests a functional contribution. Explain why the two experiments answer different questions.

6. Turn figures into Results paragraphs

6.1 Organize a paragraph around one biological question

Before writing, complete the evidence map. Then organize the paragraph around the purpose of the comparison, the principal result, any necessary supporting evidence, and the conclusion justified by those findings. The purpose is to make the argument traceable. A list of software operations is usually better placed in Methods. [2]

Use this adaptable scaffold:

> To determine whether **[process or feature]** differed between **[conditions]**, we analyzed **[measurement]** in **[biological samples]**. Compared with **[reference]**, **[condition]** showed **[direction and magnitude]**, with **[uncertainty and relevant statistical evidence]** (Fig. **[X]**). **[Additional result]** provided evidence for **[specific interpretation]**. These findings support **[bounded conclusion]**.

Do not force every paragraph into identical sentences. The paragraph should answer its question and connect logically to the next question.

6.2 Work through a numerical writing example

Illustrative teaching example only. The values below are invented to demonstrate writing and are not results from a published study.

Assume six independent cultures per group, balanced processing batches, a DESeq2 model including batch and condition, and a treatment-versus-control contrast. Suppose gene X has an RNA log2 fold change of 1.2 with a Benjamini–Hochberg-adjusted P value of 0.004. In matched proteomics samples, protein X has an estimated treatment/control abundance ratio of 1.6 with a 95% confidence interval of 1.2–2.1. No protein-activity or functional assay has been performed.

Treat protein X as a prespecified protein comparison in this example; its confidence interval is unadjusted. A proteome-wide screen would additionally require reporting the protein-level multiple-testing procedure.

> To examine the molecular response to treatment, we compared transcript and protein abundance in six independent cultures per group. Gene X showed higher transcript abundance after treatment (log2 fold change = 1.2; Benjamini–Hochberg-adjusted P = 0.004; Fig. 1A). Matched proteomic measurements also showed higher protein X abundance (treatment/control ratio = 1.6, 95% confidence interval 1.2–2.1; Fig. 1B). These measurements support a concordant increase in transcript and protein abundance under the tested conditions. They do not establish whether protein X activity changed or whether protein X contributed to the treatment phenotype.

The example names the comparison, uses measurement-specific quantities, identifies the biological replication, and limits the conclusion. The full analytical details belong in Methods. The final boundary sentence can appear here or be developed in the Discussion, depending on the manuscript's structure.

6.3 Calibrate the wording to the evidence

Overstated wording	A more defensible statement
“PCA proved that treatment reprogrammed the cells.”	“The groups differed along the displayed principal components; further analyses examined the variables associated with this pattern.”
“GSEA demonstrated pathway activation.”	“The pathway gene set was enriched toward the treatment-associated end of the ranked list.”
“The accessible peak activated gene X.”	“Accessibility increased at a region linked to gene X, supporting a regulatory hypothesis.”
“The two cell types communicated through pathway Y.”	“Expression-based analysis predicted a candidate interaction involving pathway Y.”
“The metabolite increase proved greater pathway flux.”	“The metabolite abundance increased; the contribution of synthesis, consumption, and transport remains to be tested.”
“The treatment had no effect.”	“The analysis did not provide clear evidence of a difference; the estimate and its uncertainty define the effects compatible with these data.”

Use stronger language when the design and evidence justify it. Excessive hedging can obscure a well-supported result; excessive certainty can extend it beyond its scope.

7. Write figure legends and reproducible Methods

7.1 Make a figure understandable on its own

A legend should identify the biological comparison, define every panel, explain graphical elements, report independent sample numbers, describe summaries and error bars, and state the statistical tests and adjustments relevant to those panels. Define abbreviations and image scale bars where needed.

Use this template, filling every field from the actual analysis:

> **Figure X. [A title stating the supported result.]** (A) [Measurement, biological system, and comparison.] Each point represents [unit]. (B) [Second panel and its relationship to A.] Colors

indicate [groups or values]. Data are shown as [summary] with [SD, SEM, or confidence interval, as applicable]. The analysis included [number and type of independent biological units] per group; [explain repeated or nested measurements]. Comparisons used [test or model, pairing, covariates, and sidedness where relevant], with [multiple-testing procedure]. [Define significance symbols, abbreviations, and scale bars, if used.]

Never label error bars “error” without defining them. Never report only the number of cells, images, or technical wells when the inference concerns donors, animals, or independently prepared cultures.

7.2 Report the decisions needed for reproduction

For computational work, record dataset identifiers, sample inclusion and exclusion, metadata, reference genome or database versions, software versions, preprocessing, filtering, normalization, the statistical design and contrast, multiplicity control, enrichment background, integration choices, and relevant random seeds or model splits. Explain where data, code, and analysis outputs can be obtained.

For laboratory work, report the biological material, intervention and controls, independent sample structure, primary measurements, and procedures needed to evaluate the result. A named assay or commercial kit alone rarely explains how the experiment addresses the claim.

Practice: Find one missing detail in a published legend and one missing detail in your own analysis record. Describe how each omission affects interpretation or reproducibility.

8. Assemble the manuscript

8.1 Give every section a distinct job

Component	Question it answers	Your writing task
Title	What is the main contribution?	State the biological finding or methodological contribution at the strength supported by the study.
Abstract	What was asked, done, found, and concluded?	Write a compact account of the question, design, main results, and bounded implication.
Introduction	Why does the question matter, and what remained unknown?	Move from relevant knowledge to the specific gap and study objective.
Methods	How were the evidence and comparisons produced?	Make the design, measurements, and analysis decisions inspectable and reproducible.
Results	What did the study find?	Present evidence in the order needed to answer the research questions.
Discussion	What do the findings mean, and what remains uncertain?	Evaluate the interpretation against alternatives, prior work, limitations, and useful next tests.
Data and code availability	Can another researcher inspect or reproduce the work?	Provide accurate accessions, repositories, and applicable access conditions.

Results subheadings should usually express an evidence-supported finding. For example, “Differential expression analysis” names a procedure, whereas “Treatment altered the expression of extracellular-matrix genes” communicates a result, if the data support it.

8.2 Connect sections without repeating them

The Introduction defines the problem the Results address. The Discussion explains what the evidence contributes to that problem, rather than restating every plot. Write the Abstract after the main claim and supporting results have stabilized. Manuscript organization should allow a reader to follow how each result supports the central contribution. [2]

In a multi-omics paper, connect layers through shared biological questions. Explain why the second measurement was needed, what it added, and whether it agreed with the first. Organizing the entire paper only as “RNA-seq, then proteomics, then metabolomics” can leave the biological argument unclear.

For the course mini-paper, distinguish what you reanalyzed, what the original authors experimentally demonstrated, and what you propose as future work. Experiments described in a source paper are not experiments performed by your group. Proposed work belongs in a proposal or future-directions section, not in Results.

Paraphrase published findings in your own words and cite the primary source. Preserve the original comparison and uncertainty, and distinguish the authors' proposed explanation from what their experiments directly measured. Changing a few words does not substitute for attribution.

9. Guided reading of integrated studies

Case A — Chromatin, transcription, and stem-cell function

Read: Sahinyan et al., Nature Communications (2025), DOI: 10.1038/s41467-025-62758-y. [12]

This study combines RNA-seq, ATAC-seq, histone CUT&Tag/CUT&RUN, REST occupancy profiling, genetic perturbation, and functional muscle-cell analyses. Begin with Figures 1 and 5, then compare molecular findings with the differentiation, survival, or regeneration evidence.

Your tasks: Identify what each assay measures; check which cell populations and samples support each comparison; write one observation and one integrated interpretation; select a functional experiment that changes the strength of the conclusion. Ask whether developmental effects or altered cell states could explain part of the phenotype. A histone mark should not be treated as a deterministic indicator of transcription, and measurements from different cellular contexts should not be silently merged into a single matched experiment.

Case B — Microbiome, metabolites, and host physiology

Read: Lin et al., Nature Communications (2023), DOI: 10.1038/s41467-023-42005-y. [7]

This study connects human metagenomics, targeted serum metabolite measurements, host genetic analyses, and experimental work involving bacteria, valeric acid, and bone-related outcomes.

Your tasks: Separate human associations, assumption-dependent genetic causal inference, mouse interventions, and cell experiments. Label the organism and biological scale supporting each claim. Ask which experiment would test whether the metabolite mediates the bacterial effect. Separate

bacterial-administration and metabolite-supplementation experiments should not automatically be rewritten as a combined rescue experiment or as proof of efficacy in humans.

Optional transfer case — Metabolites, chromatin, and aging

Kang et al., Nature Metabolism (2024), links SAM-related metabolic measurements, chromatin observations, and interventions in muscle stem cells. [15] Use it to compare discovery measurements, targeted confirmation, and functional recovery. Identify which observation supports each proposed connection and which alternative explanation remains.

These cases are material for critical analysis. Publication does not make every inference or reporting choice an ideal template.

10. Practice and your Week 12 writing portfolio

10.1 Practice bank

Exercise 1 — Reconstruct the argument. Select an assigned paper and write its question, competing explanations, principal claim, and most informative figure in 150–200 words.

Exercise 2 — Annotate a panel. Apply the eight questions in Section 3 to a plot. Identify one required detail that cannot be recovered from the image alone.

Exercise 3 — Repair an overclaim. Rewrite three sentences from Section 6.3. For each, specify what additional evidence would justify stronger wording.

Exercise 4 — Explain discordant omics. RNA abundance increases while protein abundance does not clearly change. Propose two biological explanations and one technical explanation. Identify a measurement that could distinguish them.

Exercise 5 — Design the next experiment. A differential ATAC-seq peak lies near a differentially expressed gene. Propose a test of the putative regulatory relationship, including the comparison, controls, readout, and alternative outcomes.

Exercise 6 — Audit replication and evaluation. A study profiles many cells from a small number of donors, then splits cells randomly into training and test sets. Identify separately the problem for between-group inference and the problem for prediction in new donors.

Exercise 7 — Write from two omics layers. Use your own verified outputs to write two connected Results paragraphs. Explain what each layer contributes and preserve any disagreement or uncertainty.

10.2 Assemble your portfolio

Prepare the following materials as formative work toward the final project:

1. **A one-page paper evidence map** using the fields in Section 1.3.
2. **Two connected Results paragraphs, approximately 250–350 words in total,** integrating at least two omics layers from your project or assigned material.
3. **One complete figure legend** for a single figure or a coherent set of panels.
4. **One short Discussion paragraph, approximately 150–200 words,** explaining the principal interpretation, an alternative explanation, and the most useful next test.

5. **A reproducibility record** listing the data source, sample structure, analysis, comparison, software information, and relevant statistical decisions.
6. **A claim audit** linking each major sentence to its figure or result table and marking any statement that is a hypothesis, a published finding, or proposed future work.

These tasks prepare the existing final project; they do not introduce a new course grading scheme.

10.3 Check your draft before Week 13

- Can another reader state your biological question after reading the first paragraph?
- Is every major claim linked to identifiable evidence?
- Are comparisons, directions, units, sample numbers, and statistical labels correct?
- Have you distinguished biological replication from repeated measurement?
- Does the wording preserve the limits of the design and the assays?
- Have you reported important discordance, null results, and alternative explanations?
- Are published results clearly distinguished from your own reanalysis and proposed experiments?
- Can you explain and verify every sentence produced with AI assistance?

Bring the evidence map and first Results draft to Week 13. Language editing should preserve the measurement, comparison, magnitude, uncertainty, and strength of each claim.

Figure and worksheet plan for the expanded handout

Use the following visual sequence to support the reading. Scientific plots should use actual cited data or explicitly labeled teaching data. The framework does not reproduce or recreate published data panels.

Item	Content	What you should do with it
Figure 1	An annotated paper showing question, evidence, and claim	Locate where the argument is established and supported.
Figure 2	A sample-design diagram linking participants or animals to omics measurements	Identify independent units and matched observations.
Figure 3	Annotated PCA, volcano plot, and heatmap	Explain axes, scales, uncertainty, and limits of interpretation.
Figure 4	RNA and protein measurements with concordant and discordant examples	Separate molecular agreement from functional evidence.
Figure 5	Regulatory-region tracks alongside transcript measurements	Distinguish accessibility, occupancy, annotation, and regulation.
Figure 6	Microbiome, metabolite, and host measurements	Identify the evidence required for each proposed connection.
Figure 7	Single-cell embedding and spatial localization	Separate cell identity, state, location, and inferred interaction.

Item	Content	What you should do with it
Figure 8	Unsupervised integration and supervised model evaluation	Distinguish shared variation, discrimination, and validation.
Figure 9	Orthogonal assay, perturbation, and rescue designs	Explain which question each experiment addresses.
Figure 10	A figure-to-paragraph example with sentence annotations	Link each sentence to a measurement or inference.
Worksheet 1	Paper evidence map	Reconstruct and evaluate a published argument.
Worksheet 2	Claim audit and reporting checklist	Prepare the first draft of your mini-paper.

Selected readings and source notes

Start with readings 1–2 and one of the two main case studies, 7 or 12. Consult the methods papers as needed when completing your evidence map. The reading sequence and worksheets above are a course-specific synthesis; they are not a reproduction of any single source.

[S1] Course syllabus. Bioinformatics: From Multi-Omics Data to Discovery, Fall 2026. Supplied course document. Pages 4–5 establish the weekly sequence, Week 12 and Week 13 topics, and the multi-omics mini-paper project.

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